

Draft - expert review pending

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Topic 4.1 - The workplace as an environment

A real classroom situation

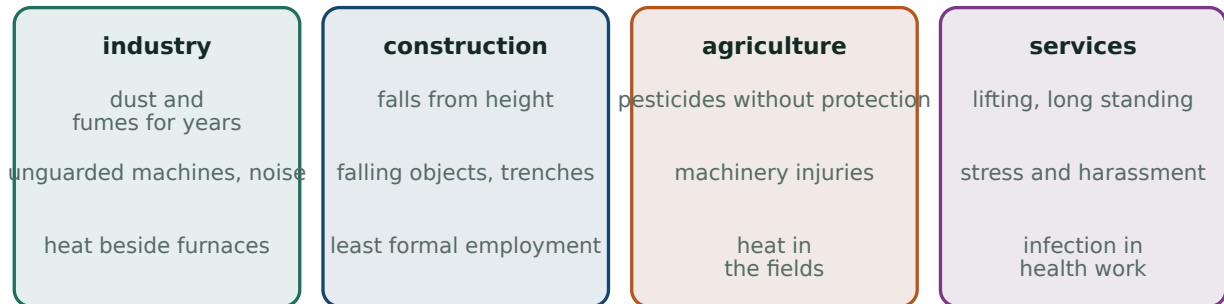
In Faisalabad, a trainee teacher's uncle has worked in a spinning mill for nineteen years. He is proud of the job and coughs a little when he says so. "Cotton gets into everyone," he says, waving at the air. "You get used to it." A study of eleven Faisalabad spinning mills found exactly what his wave hides: workers carrying respiratory disease and losing work days to it (Khan et al., 2024). His workplace, like every workplace, is an environment with its own exposures, and this topic begins there.

Explanation

Workplace environment: OSH issues in industry, construction, agriculture and services

Every sector has a characteristic hazard portrait, a mix of the five classes from Unit 3.

Four sectors, four hazard portraits



The portrait rule

Controls must match the portrait: extraction and masks answer industry's dust; guardrails and trench support answer construction; training and protection answer agriculture's chemicals; staffing and dignity answer services.

Every portrait is a mix of Unit 3's five hazard classes.

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- **Industry** (textiles, leather, steel, chemicals): dust and fumes breathed for years, unguarded machines, noise, heat beside furnaces, and chemical handling. In the Pakistani textile industry the prevalence of workplace respiratory illness, including byssinosis, cotton-dust lung disease, is high (Nafees et al., 2019).
- **Construction**: falls from height, falling objects, excavation collapse, electrical contact, and the least formal employment of any sector, which is why the Karachi site study found its hazards in excavation, unsafe practices and missing protection (Abbasi et al., 2022).
- **Agriculture**: pesticide exposure without protection, machinery injuries, heat, and the same dust burden as industry carried in a more scattered workplace.
- **Services** (shops, schools, hospitals, transport): long standing, lifting, stress, violence and harassment, and in health work, infection. The World Health Organization names mental health at work and workplace violence among its occupational concerns (WHO, n.d.).

The portrait matters because controls must match it: a dust mask programme answers industry, guardrails answer construction, pesticide training answers agriculture, and staffing and dignity answer services.

Accidents, injuries and workplace fatalities

An incident is any unplanned event that did or could have caused harm; the injury, the fatality and the near-miss are its grades. The famous case in Pakistan is the garment factory fire in Karachi in September 2012, in which more than 250 workers died, most of them trapped behind locked and barred exits, one of the deadliest industrial fires anywhere on record. Read against this unit's later topics, the case is a checklist of absent elements: no functioning alarm and evacuation, exits locked against theft, stored combustibles, no enforced housekeeping, and no management system that would have made any of these visible. The fire is extreme in its toll but ordinary in its mechanism: several small failures lining up.

Accident and injury types by sector

Sector	Characteristic accident	One documented cause	One prevention
Industry	caught-in and dust disease	output pressure, missing guards and extraction	guarding, extraction
Construction	falls and trench collapse	unsafe practices, missing protection (Abbasi et al.)	guardrails, support
Agriculture	spraying and machinery injury	unlabelled chemicals, no protective equipment	training, labels
Services	strain, stress, assault	long hours, thin staffing, no reporting channel	staffing, reporting

The quiet grades, dust disease and stress, never reach the fatality statistics that count them.

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A worked example. Take the uncle's mill and run it through the incident grades. The near-miss he does not report, a cotton bale falling short of a walkway; the injury his colleague carries, a hand caught in an unguarded roller; the fatality the mill has so far avoided; and the chronic illness, his cough, that fatality statistics never count. A workplace's real safety picture includes all four grades, and the quiet ones carry more burden than the loud ones, exactly as Unit 3's hazard teaching predicted.

Common misconception. "Accidents happen to careless workers."

Carelessness exists, but the record points elsewhere: workers enter unsafe workplaces because they need the wage, not because they misjudge the risk. A locked exit is not a worker's carelessness. The remedies that work change the workplace, not the worker's attitude.

Activity: Portrait of four sectors

Work in four groups; each takes one sector. In fifteen minutes, list at least five hazards for your sector, classify each into Unit 3's five classes, and name the one control you would demand first if your group were the safety committee. Then each group presents its portrait in ninety seconds. Allow 25 minutes in total.

Check your understanding

1. Name the four sectors this topic covers and one characteristic hazard of each. (*Remember*)
2. Why does the textile sector's hazard profile centre on dust? Use the Faisalabad and MultiTex findings. (*Understand*)
3. The 2012 Karachi fire is described as "ordinary in its mechanism". Explain what that means, using the idea of small failures lining up. (*Analyze*)
4. A farm worker sprays pesticide without protection because the employer provides none. Which control level is missing, and who holds the duty to supply it? (*Apply*)

Summary

Each sector, industry, construction, agriculture and services, carries a characteristic hazard portrait drawn from the five classes, and controls must match the portrait: extraction and masks for industry's dust, guardrails and trench support for construction, training and protection for agriculture's chemicals, staffing and dignity for services. Incidents grade from near-miss to fatality, and chronic illness is the grade statistics under-count. The 2012 Karachi

factory fire shows small failures lining up behind a locked door. Workers accept risk for the wage; the remedies that work change the workplace.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the four sectors and a characteristic hazard portrait of each.
- ☐ I can grade an event as near-miss, injury, fatality or chronic illness.
- ☐ I can read a real case, like the mill or the fire, as a checklist of absent controls.
- ☐ I can explain why "careless workers" is the wrong first explanation.

Try this at your practicum school

A school is a service workplace. Walk it as one: list five hazards the staff face, from lifting the water drum to exam-week stress, classify them, and name the one control the school's own safety committee, if it had one, should demand first. Do not name the school or any person.

Summative task

Write 160 to 200 words on the following.

Your cousin is about to join a spinning mill, a construction site or a shop, and asks which is "safest". Using the sector portraits and the incident grades, write an honest answer: what each workplace is likely to expose them to, what question they should ask the employer, and what they should look for on the first walk-through.

Mini-rubric

Criterion	What a strong response does
Portraits used (Understand)	Compares the sectors on their actual hazards, not reputation
Grades applied (Analyze)	Includes the quiet exposures, dust and stress, not only accidents
Practical advice (Apply)	Names a real question and a real first-day check
Clarity and register	Plain words for a cousin; within the word count

Further reading

- Khan, M., Muhmood, K., Mahmood, H. Z., & Khaliq, I. H. (2024). The health and economic burden of dust pollution in the textile industry of Faisalabad, Pakistan. *Journal of the Egyptian Public Health Association*, 99(1).
<https://doi.org/10.1186/s42506-024-00150-2>
- Nafees, A. A., De Matteis, S., Kadir, M. M., & Burney, P. (2019). MultiTex RCT - a multifaceted intervention package for protection against cotton dust exposure among textile workers in Pakistan: study protocol. *Trials*, 20.
<https://doi.org/10.1186/s13063-019-3743-3>
- Abbasi, H. N., Khan, R., Nasir, I., & Ahmed, W. (2022). A case study of Tariq Road Underpass, Karachi: Environmental analysis. *International Journal of Economic and Environmental Geology*, 12(4).
<https://doi.org/10.46660/ijeeg.Vol12.Iss4.2021.651>
- World Health Organization. *Occupational health* (health topic):
<https://www.who.int/health-topics/occupational-health>
- Harrington, J. M., & Gill, F. S. (Eds.). *Occupational Health*. Blackwell. (guide-recommended book, cited by reference only)

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